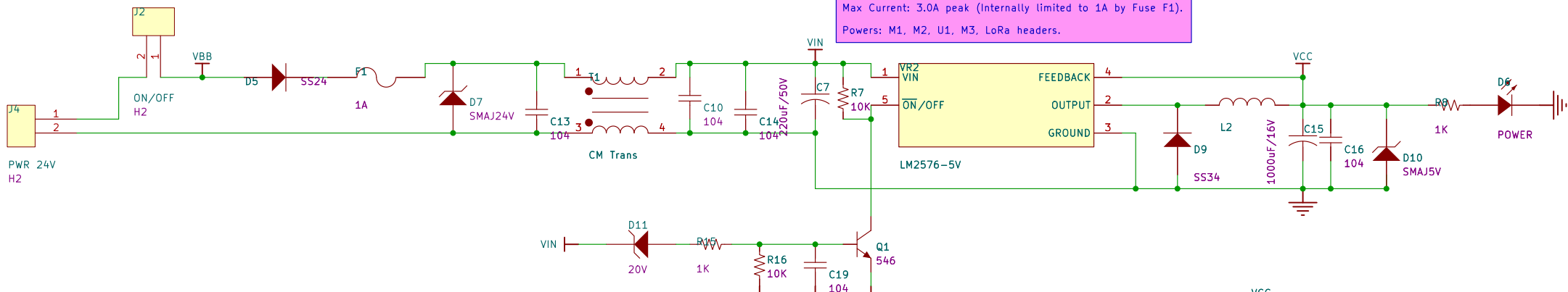


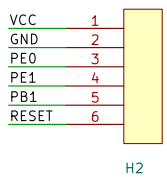
LOGIC POWER SUPPLY (VR2)

Function: Generates the core 5V rail for all logic ICs.
Input: 24V DC (Filtered via T1/F1).
Output: 5.0V DC (Fixed).
Max Current: 3.0A peak (Internally limited to 1A by Fuse F1).
Powers: M1, M2, U1, M3, LoRa headers.



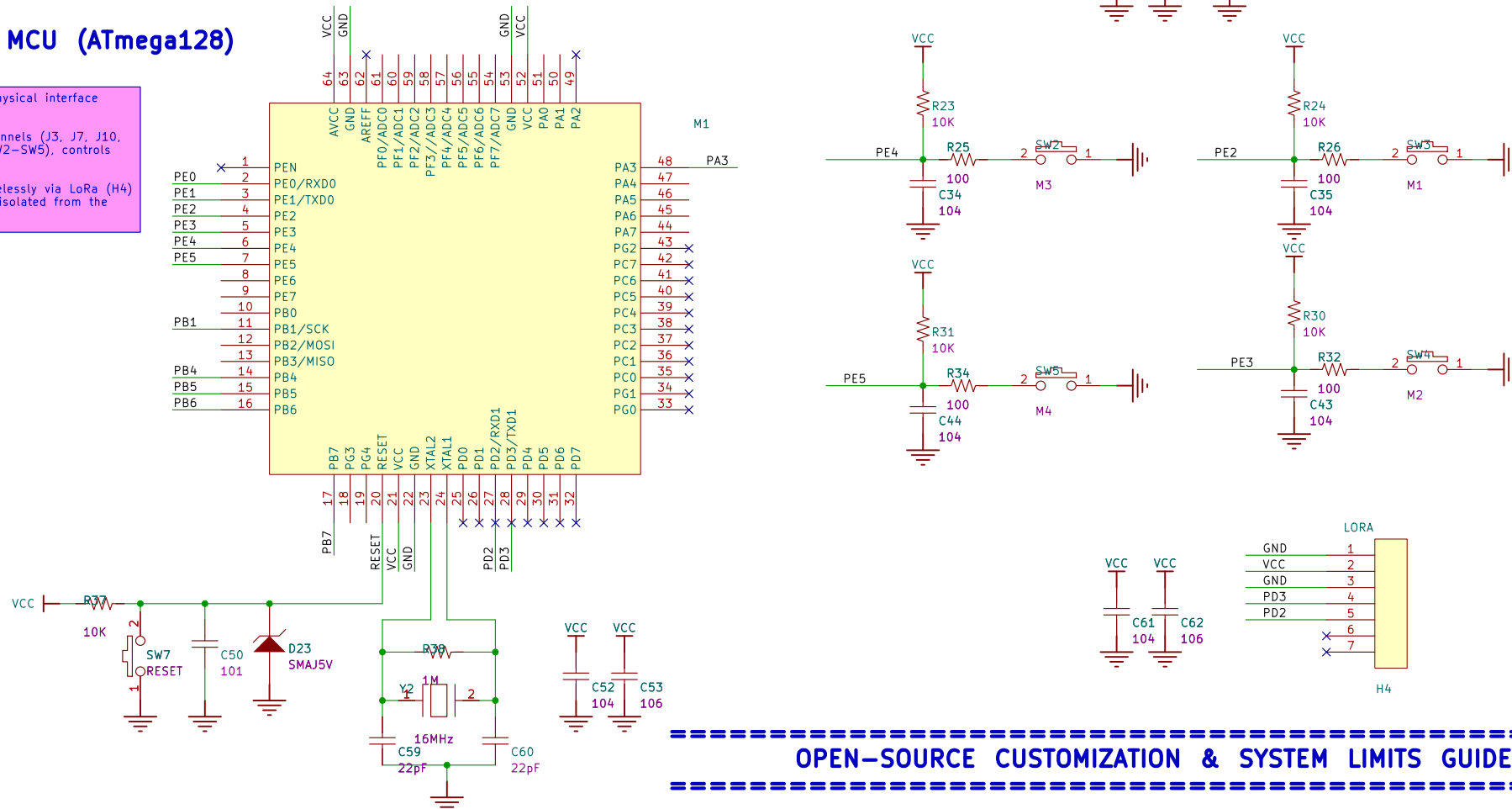
UNDER-VOLTAGE LOCKOUT / UVLO (Q1, D11)

Function: Prevents the 5V logic rail (VR2) from booting if the input voltage drops below safe operational levels, protecting M1/M2 from brown-out memory corruption.
CUSTOMIZATION: The boot threshold is physically determined by the Zener Diode D11 (Currently a 20V Zener). To adapt this board for a 12V industrial system, replace D11 with a 9V or 10V Zener diode.



M1: ACTUATION MCU (ATmega128)

Role: Dedicated hardware and physical interface controller.
Tasks: Drives 4x PWM motor channels (J3, J7, J10, J11), reads local UI buttons (SW2-SW5), controls buzzer (LS1).
Comms: Receives commands wirelessly via LoRa (H4) or wired serial (H2). Physically isolated from the RS485 network.



OPEN-SOURCE CUSTOMIZATION & SYSTEM LIMITS GUIDE

This board features a bifurcated architecture (MCU M1 = Motors, MCU M2 = Comms). Below are the hard hardware limits and the supported customization paths.

[1] HARD SYSTEM LIMITS (DO NOT EXCEED)

- * ABSOLUTE MAX INPUT VOLTAGE: -26V DC. The TVS Diode (D7) is an SMAJ24V. Sustained input voltages above 26V will cause D7 to short to ground, blowing the main fuse or the diode.
- * LOGIC CURRENT LIMIT: 1.0 Amp.
- The 5V regulator (VR2) and all logic/comms are protected by fuse F1. Do not attempt to draw heavy auxiliary 5V power from the headers.
- * MOTOR CURRENT LIMIT: 3.0 Amps per channel (12A Total).
- This is a hard silicon limit baked into the LM2576-ADJ regulators (VR1, VR3, VR4, VR5). The flyback diodes (SS34) and inductors are also strictly rated for 3A. You CANNOT increase this via firmware.

[2] POWER SUPPLY CUSTOMIZATIONS

- * ADJUSTING MOTOR VOLTAGE (Currently -8.85V): The LM2576-ADJ regulators are set via resistor dividers. To change a motor's voltage, swap its high-side (R_top) 62K resistor. Formula: $V_{out} = 1.23V * (1 + (R_{top} / 10K))$.
- * MOTOR 1 (VR1): Change R5
- * Motor 2 (VR3): Change R19
- * Motor 3 (VR4): Change R27
- * Motor 4 (VR5): Change R39

- * CHANGING THE MAIN SYSTEM TO 12V: The board is factory-locked to a 20V minimum input by the Under-Voltage Lockout (UVLO) circuit. To run the board off a 12V supply, you MUST physically replace Zener diode D11 (20V) with a 9V or 10V Zener diode.

[3] SENSOR NETWORKING (12V vs 24V RS485 SENSORS)

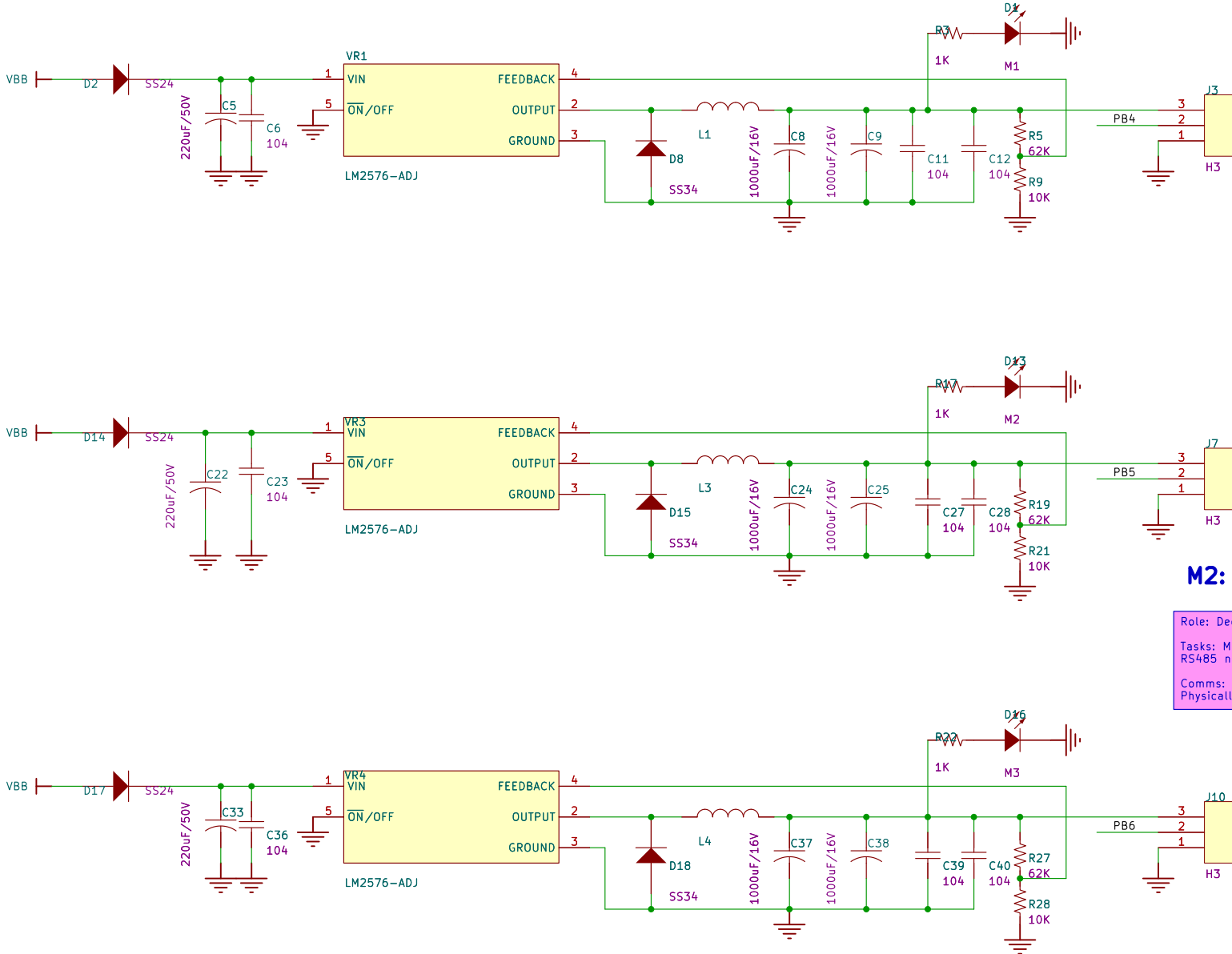
- The 4-pin RS485 sensor ports (J5, J6, J8, J9) and the DWIN port (J1) route the raw VIN power net to Pin 4 to power external nodes. There is NO dedicated 12V regulator on this board.
- * If you supply 24V to the main input (H4), the sensors receive 24V.
- * If you want to use strictly 12V RS485 sensors, you MUST: 1. Swap the D11 UVLO diode to 10V (as detailed above).
- 2. Power the entire board via a 12V main power supply. (Note: Supplying 12V main power means your motor regulators can now only output a maximum of -10.5V due to voltage dropout).

[4] MCU COMMUNICATION & ISOLATION

- M1 (Motors) and M2 (Comms) have NO physical copper data traces linking them. They are designed to communicate wirelessly via dual LoRa modules (H3 & H4).
- * To establish a wired link, you must manually run jumper wires between programming headers H1 and H2.
- * Turn SW1 OFF to disconnect M2's USB bridge and prevent data collisions if routing UART over the H1/H2 pins.

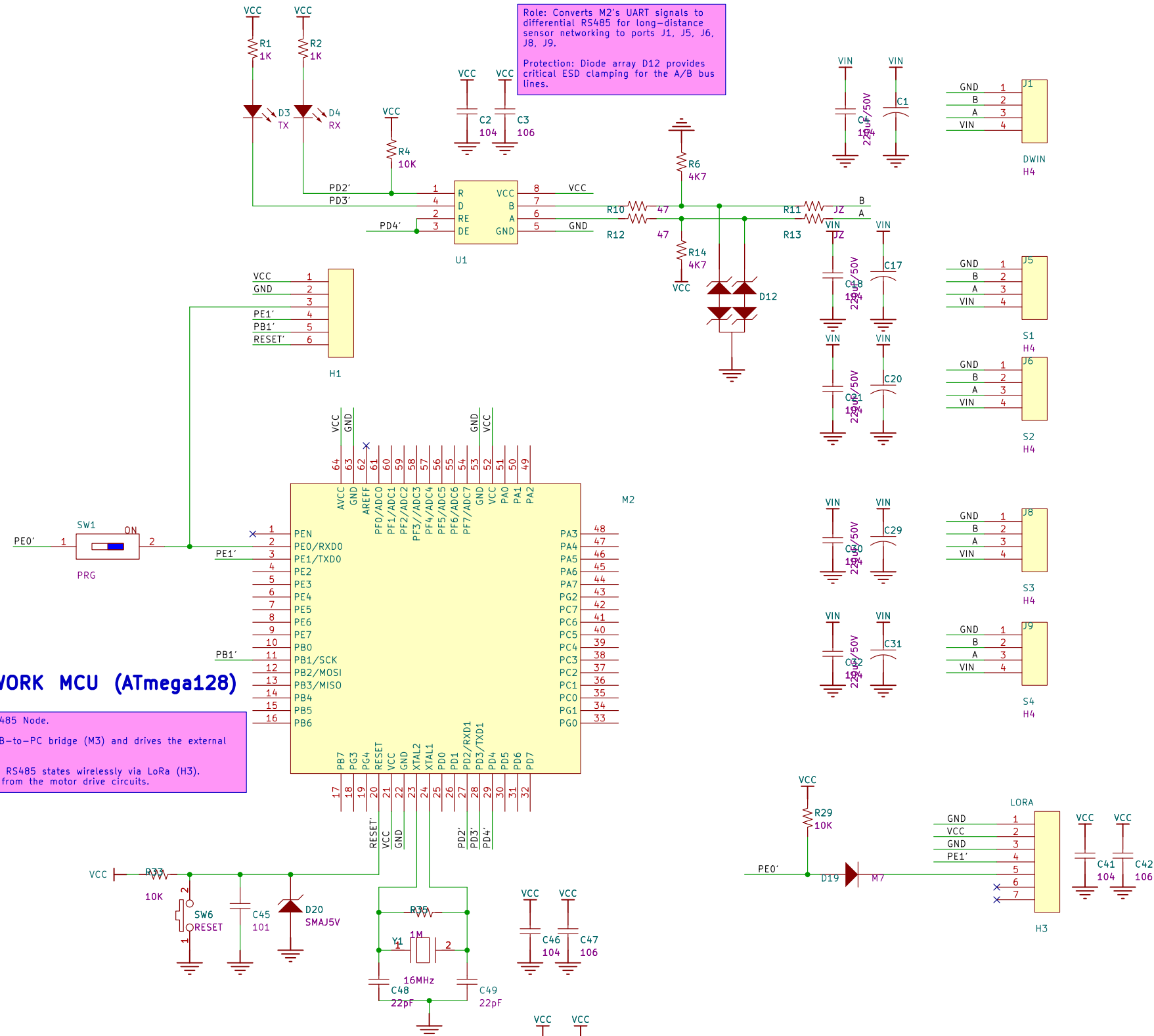
MOTOR POWER SUPPLY (CH 1-4)

Function: Independent power rails for 3-wire smart motors/servos.
Input: 24V DC (Raw VBB).
Output: -8.85V DC (Adjustable).
Max Current: 3.0A per channel (12A Total peak).
Customizability Note: Output voltage is set by R_top (62K) and R_bot (10K). To customize motor voltage, change R_top using the formula:
 $V_{out} = 1.23V * (1 + R_{top}/10K)$.



M2: NETWORK MCU (ATmega128)

Role: Dedicated RS485 Node.
Tasks: Manages USB-to-PC bridge (M3) and drives the external RS485 network.
Comms: Broadcasts RS485 states wirelessly via LoRa (H3). Physically isolated from the motor drive circuits.



AUXILIARY GPIO & SMART POWER GATING

[1] THE M2 "SMART VALVE" PIN (PB1)

- Header H1 (M2 Programming/UART) exposes MCU pin PB1 on Pin 5. This pin is electrically isolated from the core network and is completely free for custom Digital I/O logic.
- * RECOMMENDED MODIFICATION (SENSOR POWER GATING): Industrial RS485 sensors draw heavy quiescent current even when idle. Route a jumper wire from H1 Pin 5 (PB1) to the signal pin of an external 5V-triggered P-Channel MOSFET or DC Solid-State Relay (SSR). Splice the relay into the 24V line feeding the sensor headers. M2 can now toggle PB1 HIGH to wake the network, and LOW to cut power.
- * D CRITICAL SAFETY WARNING (NO N-CHANNEL / LOW-SIDE SWITCHES): If you build an external power gate, you MUST use HIGH-SIDE switching (cutting the 24V line). Do NOT use cheap N-Channel MOSFETs to cut the Ground wire. Cutting the Ground will force the 24V return current backwards through the RS485 data lines, instantly destroying U1 and M2.

[2] OTHER AVAILABLE EXTENSION PINS

- * M1 AUXILIARY I/O: Header H2 (M1 Programming/UART) exposes a mirrored free GPIO on Pin 5. This pin is electrically isolated from the core network and is completely free for custom Digital I/O logic.
- * THE HIDDEN I2C BUS (Pins 25 & 26): The ATmega128 hardware I2C pins (PD0/SCL and PD1/SDA) on both M1 and M2 were left floating (unrouted). Advanced users can micro-solder magnet wire directly to these legs, add 4.7K pull-up resistors, and establish a high-speed, collision-free internal communication bus between the Actuation and Network domains.